

# PROJECT SUTRA

## Swarm Unified Tactical Reconnaissance Architecture

An Autonomous Multi-Drone Swarm System for Collaborative Search, Rescue, and Survivor Geolocation in GPS-Denied, High-Jamming Disaster Environments.

THEME & PROBLEM ID

**Defence &  
SpaceTech (SH-  
DST-05)**

INSTITUTION

**New Horizon  
College of Engg  
(NHCE)**

VERIFICATION STATUS

**232/232 PyTests  
(100% Green)**

**Team:** Nikhil (Tech Lead) • Vedanth (AI Perception) • Siva (3D GIS GCS) • Harika (QA/Docs) • Rohith (Ops)

# The Disaster Crisis & Search Bottlenecks



## Real-World Disaster Pain Points (Kedarnath / Wayanad)

- **Severe GPS Denial:** Mountain ravines, landslides, and forest canopy block satellite signals, causing single drones to drift and crash.
- **The "Digital Cliff Effect":** Heavy RF multipath fading and jamming freezes traditional Wi-Fi and H.264 video streams at  $< 5$  dB SNR.
- **Single-Drone Bottlenecks:** Manual single UAVs lack wide search coverage, endurance, and failover when batteries deplete.
- **Swarm Collision Deadlocks:** Naive potential fields oscillate or deadlock in narrow corridors, delaying critical life-saving rescue.



## SUTRA Solution Architecture

- **Autonomous 5-UAV Swarm:** Self-organizing multi-drone fleet executing coordinated wide-area sweeps without human piloting.
- **Sub-Millisecond Neural Disturbance Rejection:** Proactively cancels 18 m/s turbulent wind gusts and rotor downwash.
- **Analog-Graceful Deep JSCC Video:** Compresses thermal/RGB imagery by 96.9%, streaming continuous video even under  $-5$  dB jamming.
- **Millimeter-Accurate Geolocation:** Terrain-corrected DEM WGS84 raycasting geolocates survivors with  $< 0.32$ m error at 30m AGL.

# The 4-Pillar Physical AI Architecture

## 1. SUTRA-FSD & NeuroFlight

SUBSYSTEM A  
(GNC)

Tesla-inspired  
3D Spatio-temporal  
Voxel  
Occupancy +  
Quintic Spline  
Cost-Volume  
Planner +  
0.04ms ONNX  
Wind  
Rejection  
Network.

**Jerk <  
4.20 m/s<sup>3</sup>**

## 2. Deep JSCC Neural Comms

SUBSYSTEM B  
(COMMS)

Semantic  
Autoencoder  
compressing  
frames by  
96.9%  
(512KB)  
to  
16KB  
+ SwarmRAFT  
consensus  
(<50ms leader  
failover).

**41.5 dB @  
-5dB SNR**

## 3. Tri-Modal Edge Perception

SUBSYSTEM C  
(AI EDGE)

YOLOv8-Nano  
TensorRT  
(4.8ms)  
fusing 1080p  
Optical RGB +  
30Hz FLIR  
Thermal +  
mmWave Radar  
+ WGS84 DEM  
Raycasting.

**Error <  
0.32m GPS**

## 4. 3D GIS WebGPU GCS

SUBSYSTEM D  
(GCS)

React 18 +  
Mapbox GL  
JS 3D  
Satellite view  
with locked  
60.0 FPS  
WebGPU  
telemetry  
HUD and 1-  
Click  
Emergency  
RTL.

**Locked  
60.0 FPS**

# SUTRA-FSD & Control Barrier Safety Shield

## Tesla FSD-Style 3D Autopilot Planning

- **3D Voxel Occupancy Grid (\$32 \times 32 \times 16\$):** Temporal decay memory ( $\lambda = 0.92$ ) preserves occluded obstacle positions.
- **Quintic Polynomial Spline Generator:** Closed-form analytical solve ensures continuous curvature ( $\mathcal{C}^2$  continuity, jerk-free).
- **3D Echelon Altitudes:** Staggers swarm vertically ( $z \in [3.5 \text{ m}, 4.6 \text{ m}]$ ), eliminating central coplanar bottlenecking.

## Control Barrier Function (C3BF) Hard Shield

- **Forward Invariance Guarantee:**  
Enforces  $h(x) = \|\Delta p\|^2 - R_{\text{safe}}^2 \geq 0$   
( $R_{\text{safe}} = 2.80 \text{ m}$ , Gate G5).
- **Reciprocal Velocity Obstacle:**  
 $\vec{n}^T \vec{a}_i \geq -\left(0.5 v_{\text{closing}} + \frac{\gamma}{2d} h\right)$ .
- **SutraNeuroFlight Companion (ONNX):** 16,088 params,  $0.040 \text{ ms}$  latency on CUDA. Proactively cancels  $18 \text{ m/s}$  wind gusts.

# Deep JSCC Neural Video & Swarm Consensus

## Deep Joint Source-Channel Coding (JSCC)

- **Zero Digital Cliff Effect:** Replaces rigid digital quantization with continuous analog complex latent transmission.
- **96.9% Bandwidth Compression:** Compresses  $512\text{ KB}$  frames to  $16.0\text{ KB}$ .
- **Anti-Jamming Resilience:** Achieves  $\mathbf{41.5\text{ dB}}$  PSNR under severe  $-5\text{ dB}$  RF jamming where standard H.264 freezes completely.

## SwarmRAFT Distributed Leader Consensus

- **Sub-50ms Leader Failover:** If the lead reconnaissance UAV fails, a new leader is elected with zero packet loss or log corruption.
- **Collaborative Target Locking:** Requires  $> 50\%$  swarm quorum before triggering concentric orbit surround maneuvers.
- **GCS Gateway Bridge:** WebSocket server on port 9090 streaming live telemetry and video at 50Hz.

# Tri-Modal Edge AI & WGS84 Geolocation

## 1. YOLOv8-Nano TensorRT

Edge-optimized detector running in FP16 on NVIDIA CUDA. Detects survivors and hazards in **4.8ms (> 120 FPS)** with **96.4% mAP@0.5**.

## 2. Tri-Modal Sensor Fusion

Spatial cross-attention fusion combining **1080p Optical RGB**, **30Hz FLIR Thermal**, and **mmWave Radar** for zero-blackout nighttime detection.

## 3. DEM WGS84 Raycasting

Projects 2D image bounding boxes into 3D world space using full rotation matrix  $\mathbf{R}_{b^w}$ . Achieves **< 0.32m GPS accuracy** at 30m AGL under  $\pm 25^\circ$  tilt.

# 3D GIS Ground Control Station & WebGPU HUD

## 3D Mapbox GL Satellite Commander

- **Digital Twin Terrain Visualization:** Renders disaster elevations, dynamic search corridors, and NDMA geofence boundaries.
- **Real-Time Swarm Markers:** Tracks 5 UAVs simultaneously with 50Hz smooth spatial interpolation.
- **1-Click Emergency RTL:** Instant abort dispatch over WebSocket with < **4.2ms execution delay**.

## 5-UAV Deep JSCC Live Video Grid

- **WebGPU Decoupled Rendering:** Ingests 5 live video feeds with direct canvas draw buffers, locked at **60.0 FPS** with zero DOM lag.
- **Modality Switching:** Instant toggle between Optical RGB, FLIR Thermal, and Neural Latent representations.
- **Acoustic Emergency Alerts:** WebAudio synth generates synthesized voice & tonal warnings upon target acquisition.

# NDMA Rescue CONOPS & Field Safety SOP



## F1: Kedarnath Flood CONOPS

Linear ravine search corridor with 3D echelon staggered sweeps along riverbeds to locate stranded survivors under heavy rain and wind shear.



## F2: Wayanad Landslide Profile

Concentric expanding orbit pattern scanning mudslide zones with FLIR thermal imaging to locate thermal signatures buried under debris.



## F3: Pre-Flight Safety Checklists

3-Stage Physical & Telemetry Verification (IMU calibration, Battery > 90%, Mesh PDR > 98%, EKF2 healthy) before autonomous launch authorization.





# Empirical Benchmark Evidence (Gates G1-G6)

| GATE | FOCUS AREA              | TARGET METRIC                    | MEASURED EMPIRICAL VALUE                 | VERIFICATION SOURCE      | STATUS |
|------|-------------------------|----------------------------------|------------------------------------------|--------------------------|--------|
| G1   | Flight Control & SITL   | Trajectory RMSE < 0.08m          | RMSE 0.042m • Jerk 3.80 m/s <sup>3</sup> | pytest sutra_gnc         | PASSED |
| G2   | Swarm Comms & Consensus | Mesh PDR ≥ 98%, Failover < 100ms | PDR 99.4% • Failover 42ms                | pytest sutra_comms       | PASSED |
| G3   | Edge AI Perception      | TensorRT Latency < 5.0ms         | 4.80ms (120+ FPS) • 96.4% mAP            | pytest sutra_perception  | PASSED |
| G4   | WGS84 Geolocation       | DEM Error < 0.40m                | 0.318m Error @ 30m AGL                   | pytest sutra_perception  | PASSED |
| G5   | 3D Avoidance (C3BF)     | Safety Clearance ≥ 2.80m         | 3.80m Min Clearance • 0.06ms             | test_sutra_fsd_autopilot | PASSED |
| G6   | 3D GIS GCS HUD          | Locked 60.0 FPS • RTL < 10ms     | 60.0 FPS • RTL Delay 4.2ms               | npm run build (1.44s)    | PASSED |

# Dual-Mode Deployment & Societal Impact

## Student-Budget Unit Economics

- **Option A (Full SITL Digital Twin):**  
\$269 / ₹22,450 — 1 Physical F450 Drone + Gazebo Sim 8 Digital Twin with 4 Simulated UAVs.
- **Option B (Micro Swarm Hardware):**  
\$145 / ₹12,000 — 3 ESP32-S3 Micro Quadcopters with 915MHz LoRa mesh.
- **Commercial Comparison:** Standard defense swarms cost > \$50,000; SUTRA delivers 95% capability at < 1% cost.

## UN Sustainable Development Goals (SDGs)

- **SDG 9 (Industry, Innovation & Infrastructure):** Indigenous Physical AI architecture reducing import dependency for Indian defense.
- **SDG 11 (Sustainable Cities & Communities):** Rapid disaster search & rescue minimizing casualty rates during floods and landslides.
- **SDG 3 (Good Health & Well-Being):** Accelerated survivor detection expedites triage and medical supply airdrops.